Modeling and Validation of a Soccer Ball's Three-Dimensional Flight to the Goal

Problem B

Abstract

This study predicts a football's goal-plane crossing and its uncertainty from an initial position and two subsequent observations. Four models are compared: gravity alone, gravity with drag, constant resultant force, and gravity with drag and Magnus force. The model incorporating gravity, drag, and Magnus force uses a constant drag coefficient and a linear lift relation. Initial velocity, spin direction, and aerodynamic parameters are inferred from early positions, followed by numerical trajectory integration.

Subsequently, two placement shots and one power shot are selected as representative cases spanning different kick intensities and initial conditions. Predictions are evaluated against subsequent measured positions; the model incorporating gravity, drag, and Magnus force yields crossing errors of 0.360, 0.146, and 0.113 m, respectively. For the given problem, the reference crossing is (11, 2.616, 1.953) m, reached at approximately 0.399 s.

Sensitivity analysis examines initial motion, aerodynamic parameters, and air density. A 1° change in launch direction shifts the crossing by approximately 0.20 m, while a 5% speed change alters arrival time by up to 0.021 s. Spin magnitude and the lift parameter jointly determine deflection through their product. Different aerodynamic and initial-motion combinations reproduce the early positions within a 2 mm coordinate tolerance, while their crossings deviate from the baseline trajectory by up to 0.1074 m within the specified aerodynamic parameter ranges. Next, independent uniform perturbations of nine parameters generate 100,000 Monte Carlo trajectories, defining a crossing distribution and a joint 95% region; 95.772% clear the goal opening.

Keywords: penalty trajectory; aerodynamic drag; Magnus effect; parameter estimation; uncertainty analysis

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1 Introduction

1.1 Problem Background

A penalty kick is a direct contest between a designated kicker and the opposing goalkeeper. The kicker strikes the ball from the penalty mark towards the goal. In a typical professional penalty, the ball then flies freely through the air towards the goal, while the goalkeeper must choose where and when to move. An effective shot depends on speed and placement, as well as the kicker's ability to conceal its intended direction.

The importance of penalties to match outcomes has motivated research into shooting and saving, particularly the ball's motion after it leaves the foot. Topics include curved trajectories and aerodynamic forces[3], surface structure[5, 4], and trajectory measurements and aerodynamic-parameter estimation[9]. These studies connect kicking technique with the physics of ball flight and help explain the motion of a penalty shot.

1.2 Problem Restatement

Problem B concerns the prediction of a penalty trajectory from its early flight positions. The goal is 11 m from the penalty mark, 7.32 m wide and 2.44 m high. The task is to develop and test a general method that uses the ball-centre coordinates at 0.05 s and 0.10 s after the kick to predict its most likely entry point and estimate the uncertainty of that prediction.

The preceding discussion suggests three broad stages of the ball's motion: **contact with the kicker**, **free airborne flight** and **motion after the shot**. We examine the free flight after ball–foot contact has ended and before the ball enters the goal; the effect of contact is represented by the initial velocity and spin. The forces and governing equations are discussed further in Section 3. Initial-state inference from early positions, subsequent trajectory prediction and assessment of those predictions are discussed further in Section 4. Section 5 uses the specified example to examine parameter sensitivity and the distribution of goal-plane crossings under stated perturbations, connecting the predicted location with its uncertainty. The technical workflow is summarized in Figure 1.

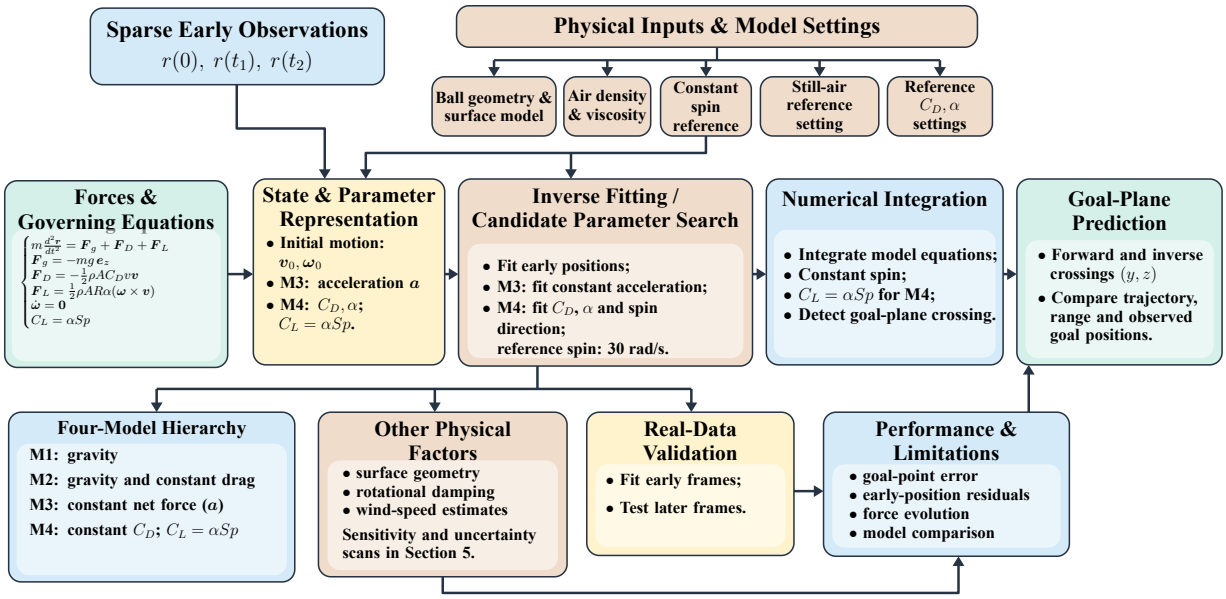


Figure 1: Three-force modeling, inverse prediction and real-penalty validation.

2 Preparation of the Models

2.1 Assumptions

1. **Free flight.** At the start of the motion considered, the ball has left the kicker's foot; contact is represented by the initial velocity and spin. We assume a proficient kick, with the ball remaining airborne without contacting the ground or another object before reaching the goal plane.

2. **Rigid spherical ball.** The ball remains spherical and its mass is uniformly distributed. Seams and asymmetric surface geometry are discussed further in Section 6.1.

3. **Uniform gravity and air properties.** The magnitude and direction of gravitational acceleration remain constant over the flight, and air density and viscosity remain uniform.

4. **Still air.** The surrounding air is treated as stationary. The effects of wind on relative airflow and aerodynamic force are discussed further in Section 6.3.

5. **Constant spin.** Aerodynamic torque is neglected, so the spin vector remains constant during flight. Rotational damping is discussed further in Section 6.2.

2.2 Notations

The symbols used below and their physical meanings are summarized in Table 1.

Table 1: Principal quantities and competition reference values.

Symbol	Meaning	Value or unit
$\boldsymbol{r}, \boldsymbol{v}$	Ball-centre position and velocity	m; m/s
$\boldsymbol{\omega}, \omega$	Spin vector and its magnitude	rad/s
m, R, A	Mass, radius and frontal area	0.43 kg; 0.11 m; πR^2
g	Gravitational acceleration	9.81 m/s^2
ρ	Air density	1.2 kg/m^3
$\nu, \mu = \rho\nu$	Kinematic and dynamic viscosity	1.50×10^{-5} m^2/s; Pa s
Re, Sp	Reynolds number and spin parameter	dimensionless
C_D, C_L, α	Drag, lift and linear-spin parameters	dimensionless
$\lambda_D, \lambda_\alpha$	Coefficient ratios to reference values	dimensionless
$\theta, \boldsymbol{e}_L$	Spin–airflow angle and Magnus direction	rad; unit vector
q, v	Dynamic pressure and ball speed	Pa; m/s
$\boldsymbol{Q}, I_b$	Damping torque and moment of inertia	N m; kg m^2

We follow Maponi and Piombin [3] for the ball parameters and Asai et al. [5] for the reference air properties, with dynamic viscosity $\mu = \rho\nu$.

3 Model Establishment

3.1 Geometric Relations

As shown in Figure 2, the local origin is the penalty mark, with x towards the goal, y horizontal and lateral, and z vertically upward. The goal plane is $x = 11$ m and the opening is $-3.66 \le y \le 3.66$ m, $0 \le z \le 2.44$ m.

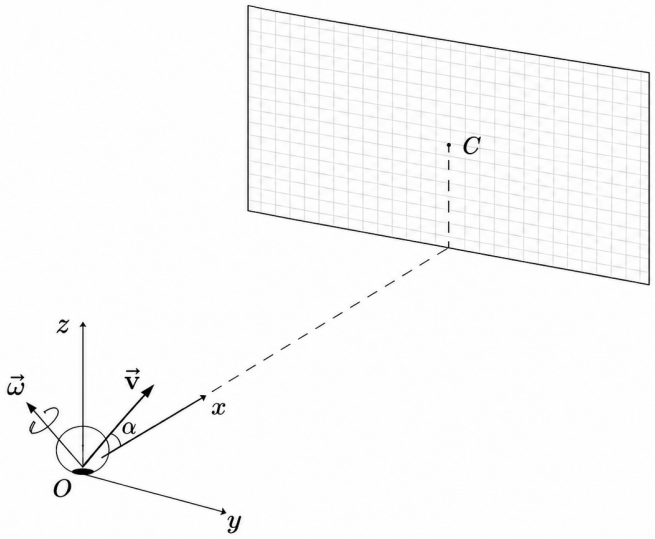


Figure 2: Penalty-kick geometry in local coordinates; O is the nominal penalty mark and C an illustrative goal-plane point.

The entire ball clears the opening when its centre satisfies $|y| \leq 3.66 - R$ and $R \leq z \leq 2.44 - R$.

3.2 Force Analysis

As shown in Figure 3, we consider gravity, buoyancy, drag and Magnus force in turn.

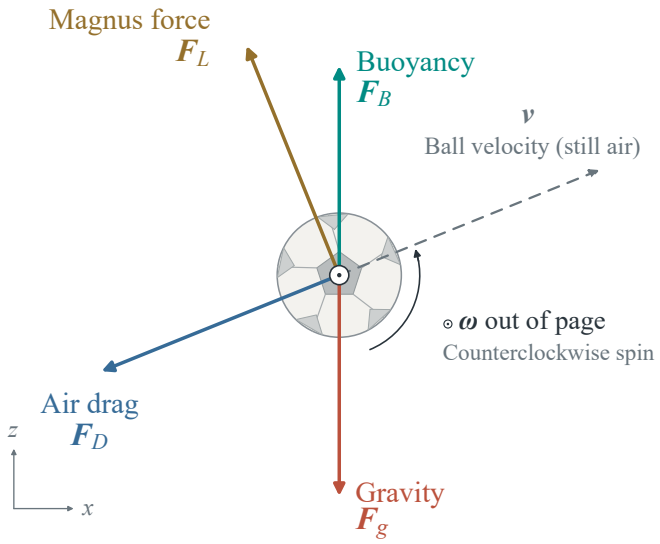


Figure 3: Schematic of gravity, buoyancy, drag and Magnus force on the football.

3.2.1 Gravity

Let $\boldsymbol{e}_z$ be the upward vertical unit vector. For mass m in the uniform field of gravitational acceleration magnitude g,

$$\boldsymbol{F}_g = -mg\boldsymbol{e}_z, \tag{1}$$

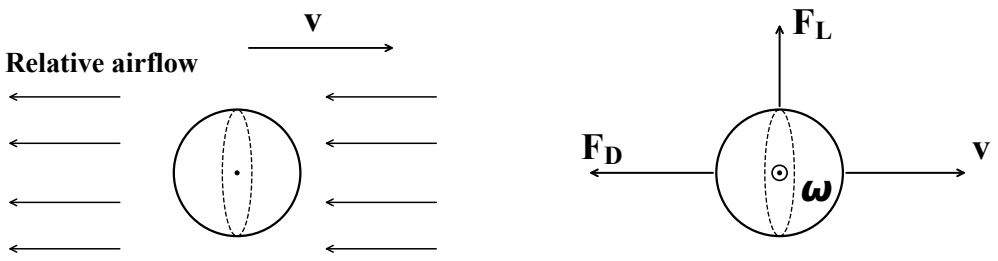
By Archimedes' principle, the displaced air also exerts an upward buoyancy force $\boldsymbol{F}_B$. For ball volume $V_b = \frac{4}{3}\pi R^3$ and air density ρ,

$$\boldsymbol{F}_B = \rho V_b g\, \boldsymbol{e}_z = \frac{4}{3}\pi R^3 \rho g\, \boldsymbol{e}_z. \tag{2}$$

From these expressions, the combined vertical force is $\boldsymbol{F}_g + \boldsymbol{F}_B = -(m - \rho V_b)g\boldsymbol{e}_z$. Buoyancy is 1.56% of weight for the reference parameters and is therefore omitted as a small correction in the subsequent calculations.

3.2.2 Aerodynamic Drag

Let $\boldsymbol{v}$ be the ball-centre velocity. Since the air is assumed stationary, the incident airflow in the ball frame has velocity $-\boldsymbol{v}$. As shown in Figure 4, the airflow slows near the forward face of the ball.



(a) Airflow in the ball frame **(b) Example: spin out of the page**

Figure 4: Schematic of relative airflow and force directions.

Let p_∞ be the incident static pressure and p_s the pressure at the forward stagnation point. Bernoulli's relation along an idealized incident streamline is

$$p_\infty + \frac{1}{2}\rho v^2 = p_s.$$

The pressure rise defines the dynamic pressure q:

$$q = p_s - p_\infty = \frac{1}{2}\rho v^2.$$

Following Goff and Carré [9], aerodynamic drag depends on air density, relative speed, frontal area and the ball's flow and surface characteristics. The latter enter through the drag coefficient C_D, so the drag magnitude is

$$F_D = C_D q A.$$

Since drag opposes the relative motion, substitution of the dynamic pressure leads to

$$\boldsymbol{F}_D = -\frac{1}{2}\rho A C_D v \boldsymbol{v}. \tag{3}$$

Here $A = \pi R^2$ is the frontal area and ρ is air density. We approximate C_D as constant during the ball's flight.

3.2.3 Magnus Lift

A spinning football changes the surrounding airflow and experiences a transverse Magnus force. As shown in Figure 5, rotation makes the relative flow asymmetric. Let $\boldsymbol{\omega}$ be the angular-velocity vector.

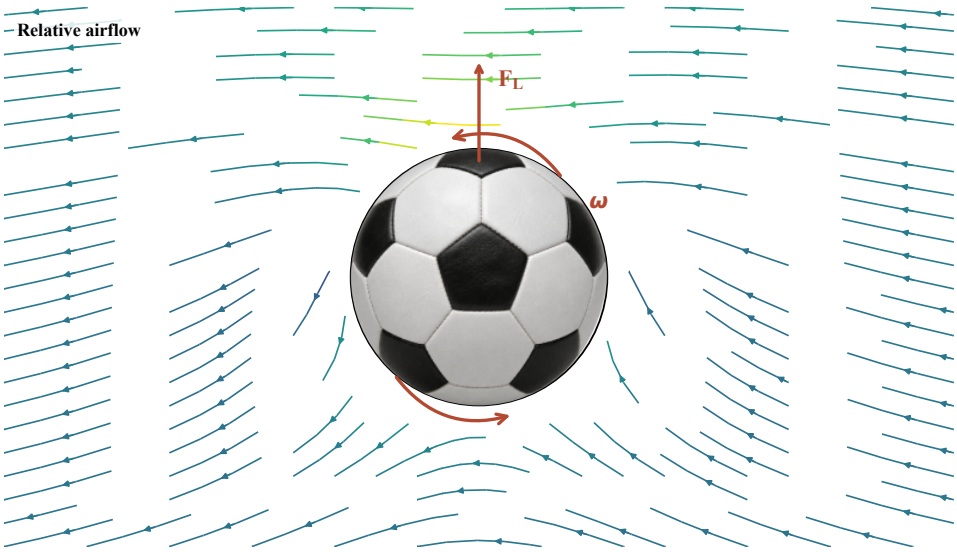


Figure 5: Idealized flow around a spinning football.

Following Goff and Carré [9], transverse lift is expressed in terms of dynamic pressure and frontal area. When the spin axis makes angle θ with the relative velocity, the lift magnitude is

$$F_L = qAC_L \sin\theta.$$

Here C_L is the transverse lift coefficient and θ is the spin–airflow angle. We approximate the relation between lift coefficient and spin parameter as linear:

$$C_L = \alpha \frac{R\omega}{v}.$$

Here α is dimensionless. Based on Kiratidis and Leinweber [6], we approximate this coefficient as constant during the flight. Substituting the dynamic-pressure and linear-spin relations,

$$\boldsymbol{F}_L = \frac{1}{2}\rho A R \alpha (\boldsymbol{\omega} \times \boldsymbol{v}). \tag{4}$$

3.3 Dynamical Equations

Since the football is treated as a rigid body, its motion can be separated into translation of the centre of mass and rotation about that centre. Let $\boldsymbol{L}$ be angular momentum about the centre, I_b the moment of inertia, and $\boldsymbol{Q}$ the resultant external torque. Centre-of-mass translation follows Newton's second law,

$$m\dot{\boldsymbol{v}} = \sum \boldsymbol{F} \tag{5}$$

while rotation about the centre follows the angular-momentum theorem,

$$\frac{d\boldsymbol{L}}{dt} = \boldsymbol{Q}, \qquad \boldsymbol{L} = I_b\boldsymbol{\omega}. \tag{6}$$

From zero applied torque and constant spherical inertia, $I_b\dot{\boldsymbol{\omega}} = \boldsymbol{0}$, so the spin remains constant. Combining this result with the three retained forces,

$$\begin{cases} \dot{\boldsymbol{r}} = \boldsymbol{v}, \\ m\dot{\boldsymbol{v}} = \boldsymbol{F}_g + \boldsymbol{F}_D + \boldsymbol{F}_L, \\ \dot{\boldsymbol{\omega}} = \boldsymbol{0}. \end{cases} \tag{7}$$

Let $\boldsymbol{r}_0$, $\boldsymbol{v}_0$ and $\boldsymbol{\omega}_0$ be the initial centre position, velocity and angular velocity, respectively; their subscripts x, y, z denote components in the chosen axes. The initial conditions are

$$\begin{cases} \boldsymbol{r}(0) = \boldsymbol{r}_0, \\ \boldsymbol{v}(0) = (v_{x0}, v_{y0}, v_{z0}), \\ \boldsymbol{\omega}(0) = (\omega_{x0}, \omega_{y0}, \omega_{z0}). \end{cases} \tag{8}$$

4 Model Solving

Early positions from three actual penalties determine inverse states whose continuations are compared with later observations. Two placement shots and one power shot provide different speeds, curvature directions and flight heights. As shown in Table 2, the four descriptions distinguish gravitational fall, drag-induced slowing, constant acceleration and spin-induced curvature.

Table 2: Four-model hierarchy and inverse quantities.

Model	Force or motion assumption	Inverse quantities
M1	gravity	initial velocity (3)
M2	gravity and drag	velocity and drag coefficient (4)
M3	constant resultant force	velocity and acceleration (6)
M4	gravity, drag and Magnus force	velocity, spin direction, C_D, α (7)

The M3 resultant includes gravity and the other forces together. In M4, C_D and α remain constant during a flight, while $C_L = \alpha Sp$ changes with speed. According to the coefficient relation of Kiratidis and Leinweber [6], we describe the flow regime by the Reynolds number $Re = \dfrac{2Rv}{\nu}$,

where ν is kinematic viscosity, and evaluate the reference coefficients at $Re^* = 333793$ and $Sp^* = 0.19$. The superscript $*$ denotes this reference condition: $C_D^* = 0.2415$ and $\alpha^* = \dfrac{C_L(Re^*, Sp^*)}{Sp^*} = 1.5251$. These values anchor the constant-coefficient approximation.

4.1 Inverse Prediction from Three Real Penalties

4.1.1 Observations and numerical method

We use the Sportec Open DFL dataset described by Bassek et al. [1] and released by DFL [2]. It combines match events with 25 Hz multi-camera ball tracking. As shown in Table 10, the three selected events comprise two placement shots and one power shot. For each flight, the first selected position after ball–foot separation defines a new time origin; the three inputs occur at $t = 0, 0.04, 0.08$ s. As shown in Table 11, their actual starting positions are retained, and the measured goal crossing supplies the later comparison. The penalty mark is the horizontal origin, with x towards the goal, y lateral and z the centre height. The goal plane is $x = 11$ m. The source records, observed positions and parameter ranges are collected in Appendix A.

Integration uses DOP853 with a maximum step of 0.005 s, relative tolerance 10^{-10} and absolute tolerance 10^{-12}. Forward roots of $x - 11 = 0$ and descending roots of $z - R = 0$ locate the goal crossing and first ground contact.

4.1.2 Inversion and reference states

M1 fits velocity, while M2 fits velocity and drag coefficient. For M3, let $\boldsymbol{a}$ be the constant acceleration in m/s^2. The three positions satisfy

$$\boldsymbol{r}(t) = \boldsymbol{r}_0 + \boldsymbol{v}_0 t + \frac{1}{2}\boldsymbol{a}t^2. \tag{9}$$

M4 fits three velocity components, two spin-direction angles and two coefficients by minimizing squared coordinate errors at the two later positions. Six deterministic starts are optimized and the smallest-residual result is selected. As shown in Table 12, the search intervals specify the allowed velocity, spin direction and coefficient ratios. Since $\alpha\boldsymbol{\omega}$ determines rotational deflection, we use a 30 rad/s reference magnitude and fit direction and α. Define the unit spin axis $\boldsymbol{n} = \dfrac{\boldsymbol{\omega}}{\|\boldsymbol{\omega}\|}$, with components n_x, n_y, n_z. As shown in Table 13, the fitted speed, coefficients and axis distinguish the initial states of the three shots.

Let $\boldsymbol{r}_i$ be a measured centre position at time t_i and $\widehat{\boldsymbol{r}}(t_i)$ its prediction; n is the number of later positions compared. Hats on the goal coordinates similarly distinguish predictions from observations. The three-dimensional root-mean-square path error RMSE_{3D} and goal-plane error E_g, both in metres, are

$$\begin{aligned} \mathrm{RMSE}_{3D} &= \sqrt{\frac{1}{n}\sum_{i=1}^{n} \|\widehat{\boldsymbol{r}}(t_i) - \boldsymbol{r}_i\|^2}, \\ E_g &= \sqrt{(\widehat{y}_g - y_g)^2 + (\widehat{z}_g - z_g)^2}. \end{aligned} \tag{10}$$

Path comparisons use observations after the fitted times and before first contact or the goal crossing, with sample count n listed in the table. E_g is evaluated when a model crosses the goal plane; its observed counterpart comes from a source frame or interpolation between frames.

4.2 Case Analysis and Validation

As shown in Figure 6, the photographs illustrate the two shot styles: Figure 6a shows the placement shot close to the turf, while Figure 6b shows the power shot higher in the air.



(a) Illustrative placement shot.



(b) Illustrative power shot.

Figure 6: Photographic examples of placement and power shots.

As shown in Figure 7, the three spatial paths differ in height and curvature. Open circles are inverse inputs, black points and lines are observations, coloured lines are predictions, and crosses are observed goal crossings.

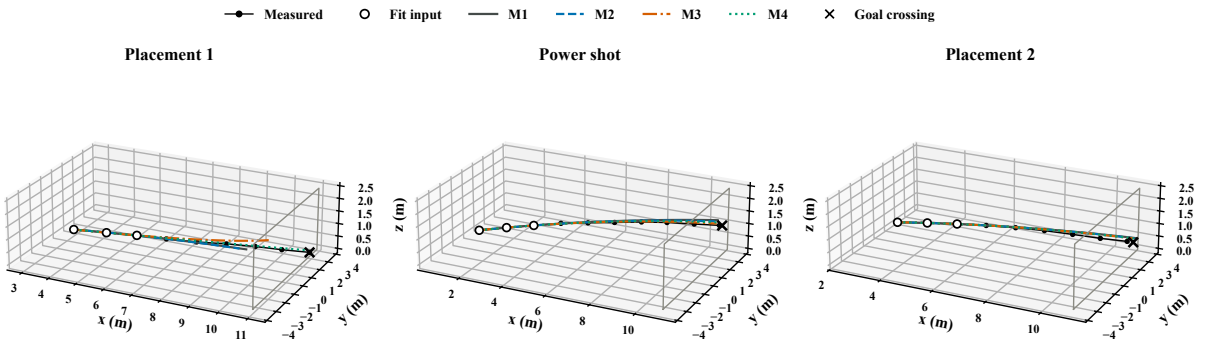


Figure 7: Observed and predicted paths for placement 1, the power shot and placement 2, from left to right. Fitting times are 0, 0.04 and 0.08 s in each case.

4.2.1 Case 1: Placement shots

As shown in Figure 8, the four descriptions produce different continuations of the two low flights.

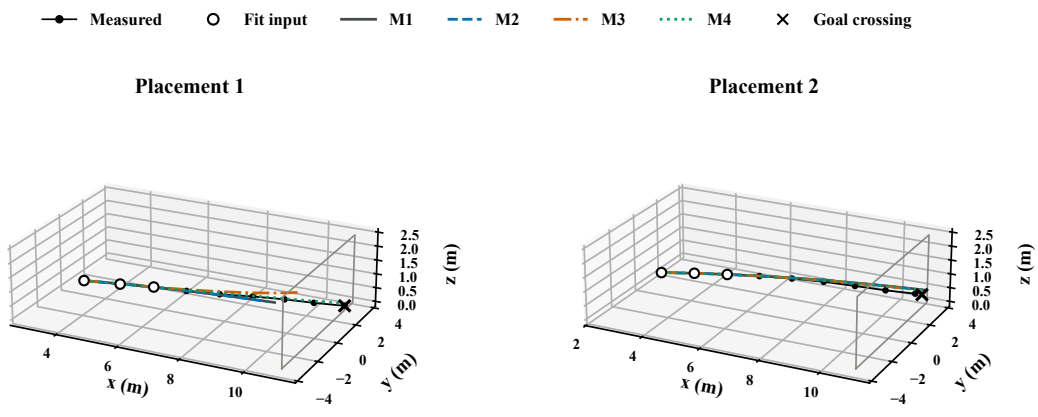


Figure 8: Measured and predicted trajectories for the two placement cases.

Both placement shots travel close to the ground. From the M4 initial states, the starting speeds are 29.20 m/s and 28.34 m/s. For placement 1, M1 and M2 reach the ground before the goal. For M3, the longitudinal acceleration is $a_x = -56.25$ m/s^2; from Eq. (9), the maximum forward position is $x = 10.973$ m, short of the goal. M4 reaches the plane with a crossing error of 0.360 m and a subsequent path RMSE of 0.234 m.

All four models reach the goal for placement 2, with crossing errors 0.475, 0.475, 0.155 and 0.146 m. Constant acceleration carries forward the early curvature in this case; M4 reduces the path RMSE from M3's 0.151 m to 0.132 m. Gravity draws a low flight downward, while the vertical Magnus component raises or lowers it according to spin direction. Together they determine whether the ball reaches the goal before contact.

4.2.2 Case 2: Power shot

From the M4 initial state, the starting speed is 32.05 m/s. As shown in Figure 9, the spatial path and coordinate histories connect forward deceleration with lateral and vertical curvature.

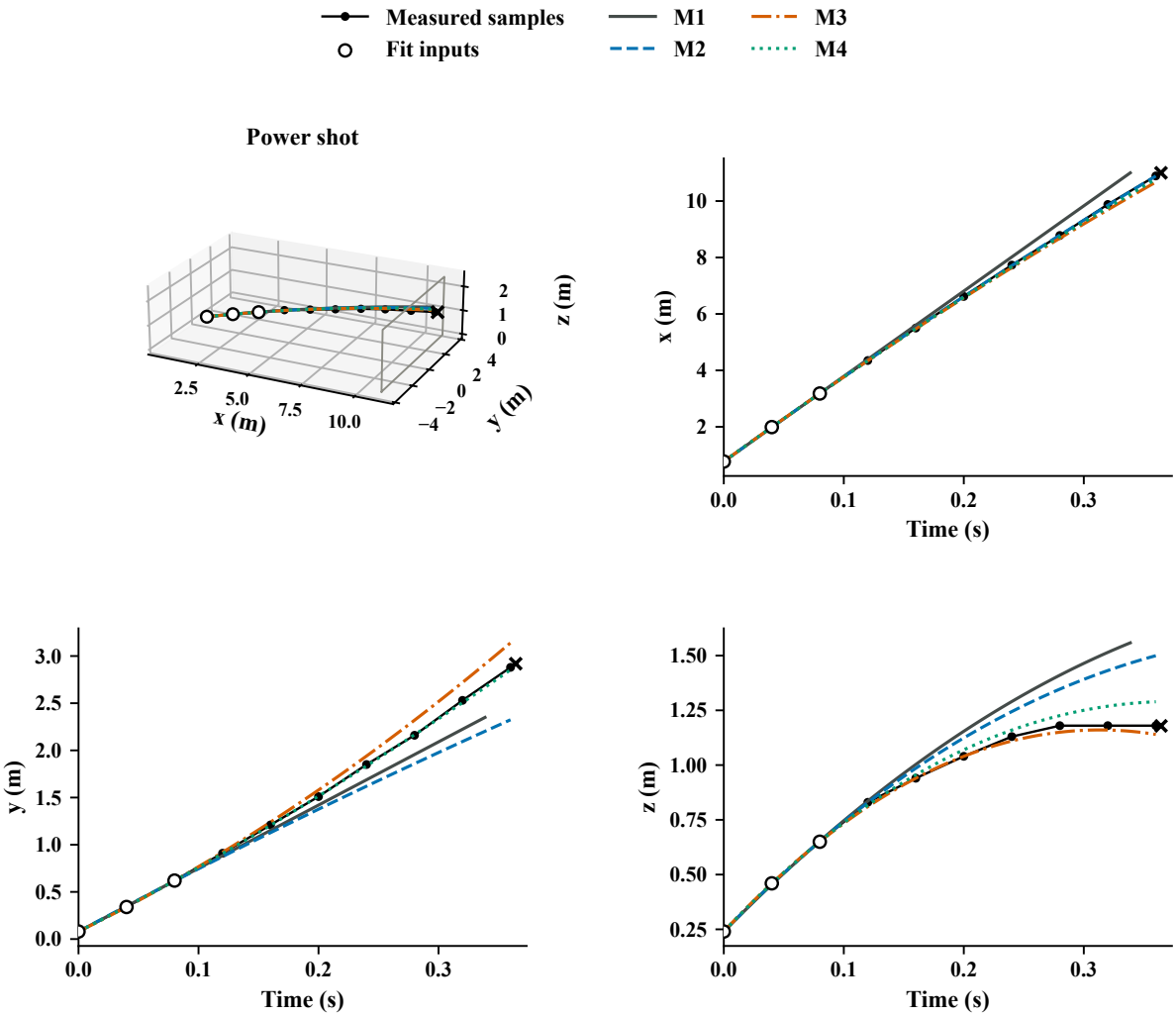


Figure 9: Power-shot path and coordinate histories. Open circles are inverse inputs, black points are observations, and coloured curves are predictions.

M1–M4 have crossing errors of 0.684, 0.657, 0.374 and 0.113 m, and subsequent path RMSE values of 0.412, 0.356, 0.190 and 0.093 m. Lateral motion in M1 and M2 follows the initial velocity; M3 extends constant acceleration. M4 curves the path through changing drag and Magnus forces, giving the smallest values of both errors for this shot.

Speed and aerodynamic forces. As shown in Figure 10, the M4 speed and force histories explain how the fitted states develop during flight. Initial drag is 7.58 N for the power shot and 7.05 N and 2.21 N for the placement shots. Drag depends on $C_D v^2$, combining speed with the fitted coefficient, and decreases as the flight slows. Linear lift depends on $\alpha(\boldsymbol{\omega} \times \boldsymbol{v})$; the angle and direction between spin and velocity determine its lateral and vertical effects.

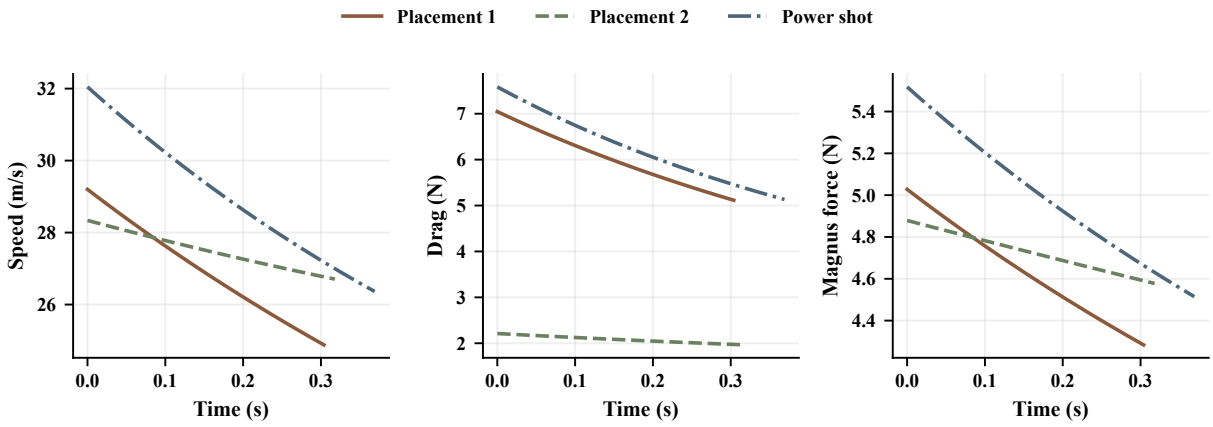


Figure 10: M4 speeds, drag and Magnus force for the three penalties. Curves end at first ground contact or the goal plane.

As shown in Table 3, early fitting residuals, subsequent path errors and goal-plane errors describe different aspects of prediction. The termination status identifies shots that reach the ground or fail to reach the goal plane.

Table 3: Early maximum coordinate residual, subsequent path RMSE/sample count, goal-plane error and termination.

Event	Model	Residual (mm)	RMSE (m)/n	Goal (m)	Status
Placement 1	M1	36.00	0.244/3	—	ground
Placement 1	M2	26.13	0.136/3	—	ground
Placement 1	M3	0.00	0.826/6	—	no crossing
Placement 1	M4	25.90	0.234/5	0.360	goal
Power shot	M1	12.00	0.412/6	0.684	goal
Power shot	M2	10.28	0.356/7	0.657	goal
Power shot	M3	0.00	0.190/7	0.374	goal
Power shot	M4	2.90	0.093/7	0.113	goal
Placement 2	M1	8.00	0.300/5	0.475	goal
Placement 2	M2	7.01	0.320/6	0.475	goal
Placement 2	M3	0.00	0.151/5	0.155	goal
Placement 2	M4	1.12	0.132/5	0.146	goal

5 Sensitivity Analysis

This chapter uses the three early positions supplied in the problem to examine how initial motion, aerodynamic parameters and air density affect the goal-plane crossing and arrival time. The model includes gravity, drag and Magnus force, with still air, constant spin, a fixed mass of 0.43 kg and a fixed radius of 0.11 m.

Table 4: Early positions for the supplied example

Time t (s)	x (m)	y (m)	z (m)
0	0	0	0.11
0.05	1.50	0.45	0.44
0.10	2.96	0.86	0.74

Applying the inverse method of Section 4 with a reference spin magnitude of 30 rad/s gives the initial state

$$\begin{aligned} \boldsymbol{v}_0 &= (30.3784,\ 9.4074,\ 6.8974)\ \mathrm{m/s}, \\ \boldsymbol{\omega}_0 &= (19.7434,\ 1.2060,\ -22.5553)\ \mathrm{rad/s}. \end{aligned} \tag{11}$$

The corresponding drag coefficient is $C_D = 0.3623$ and the linear lift parameter is $\alpha = 2.2877$. The maximum early-coordinate residual is 0.784 mm. The trajectory reaches the goal plane at $t_g^* = 0.398967$ s, with crossing coordinates $(y_g^*, z_g^*) = (2.6159, 1.9526)$ m.

Taking this trajectory as the reference, we first vary individual parameters to identify their effects on position and arrival time. We then retain the early-position constraints and examine predictions from different parameter combinations. Finally, joint perturbations provide the crossing distribution and its statistical ranges.

5.1 Effects of Individual Parameters on Crossing Position and Arrival Time

Each parameter is varied separately while the others remain at their reference values. Table 5 lists the nine perturbation ranges: percentages denote changes relative to the reference value, and direction angles vary about their reference orientation. Each parameter takes 41 equally spaced values, giving 369 trajectories.

Table 5: Perturbation ranges for the nine parameters

Parameter	Perturbation range
Launch speed	$\pm 5\%$
Velocity azimuth	$\pm 2^\circ$
Velocity elevation	$\pm 2^\circ$
Spin magnitude	$\pm 10\%$
Spin-axis azimuth	$\pm 2^\circ$
Spin-axis elevation	$\pm 2^\circ$
Drag coefficient	$\pm 10\%$
Lift parameter	$\pm 10\%$
Air density	$\pm 10\%$

The changes in crossing coordinates and arrival time relative to the reference trajectory are defined as

$$\begin{aligned} \Delta y_g &= y_g - y_g^*, \qquad \Delta z_g = z_g - z_g^*, \qquad \Delta t_g = t_g - t_g^*, \\ D_g &= \sqrt{(\Delta y_g)^2 + (\Delta z_g)^2}. \end{aligned} \tag{12}$$

Figures 11 and 12 show the crossing-position and arrival-time responses. Table 6 takes direction perturbations of $\pm 1^\circ$ and other perturbations of $\pm 5\%$ from these scans, reporting the larger displacement and absolute time change from each positive–negative pair.

Table 6: One-parameter effects on goal-plane crossing and arrival time

Parameter	Perturbation	Maximum D_g (cm)	Maximum $\lvert\Delta t_g\rvert$ (ms)
Launch speed	$\pm 5\%$	8.356	20.659
Velocity azimuth	$\pm 1^\circ$	20.070	1.965
Velocity elevation	$\pm 1^\circ$	20.218	1.692
Spin magnitude	$\pm 5\%$	3.873	0.287
Spin-axis azimuth	$\pm 1^\circ$	1.002	0.071
Spin-axis elevation	$\pm 1^\circ$	0.852	0.084
Drag coefficient	$\pm 5\%$	0.620	2.274
Lift parameter	$\pm 5\%$	3.873	0.287
Air density	$\pm 5\%$	4.142	1.990

For the perturbation sizes in Table 6, initial direction primarily affects crossing position: a 1° change in azimuth or elevation moves the crossing by about 0.20 m. Speed strongly affects arrival time; a $\pm 5\%$ change produces a maximum time difference of 20.659 ms and a maximum crossing displacement of 8.356 cm. Direction directly determines lateral and vertical motion, while speed changes the time to the goal and therefore the accumulated effects of gravity and aerodynamic forces.

The responses to spin magnitude and the lift parameter coincide because the linear Magnus force is proportional to the product $\alpha\boldsymbol{\omega}$. The same relative change in either spin magnitude or α produces the same force and trajectory response. Scaling them inversely gives

$$\boldsymbol{\omega}_0' = k\boldsymbol{\omega}_0, \qquad \alpha' = \alpha/k \quad \Longrightarrow \quad \boldsymbol{F}_L' = \boldsymbol{F}_L, \qquad k > 0. \tag{13}$$

Different spin magnitudes and lift parameters can therefore produce the same trajectory. Spin-axis orientation affects the magnitude and direction of deflection through $\boldsymbol{\omega} \times \boldsymbol{v}$. Air density scales both drag and Magnus force, changing deceleration and curvature together.

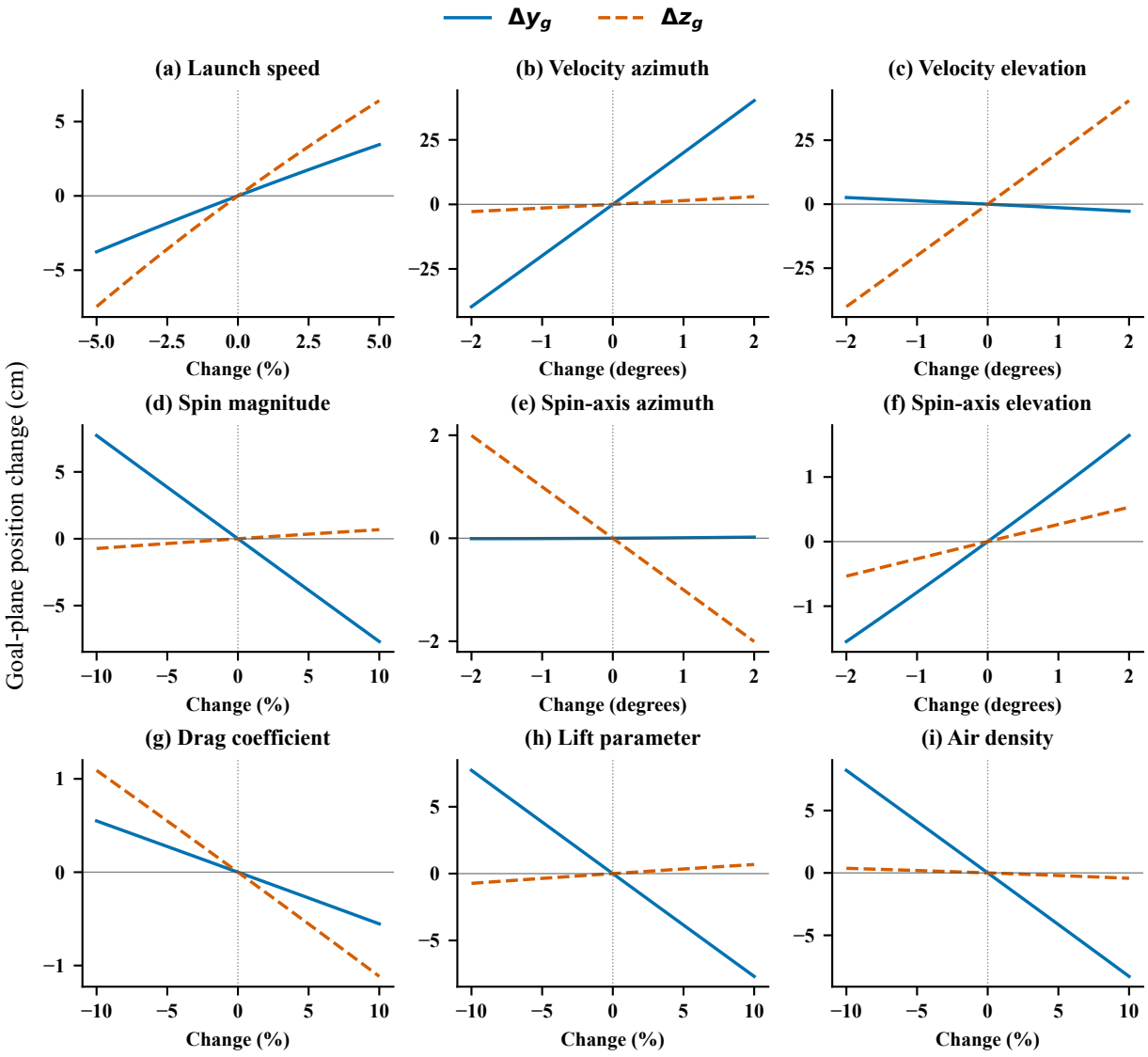


Figure 11: Goal-plane coordinate responses to nine one-parameter perturbations. Blue solid curves show Δy_g and orange dashed curves show Δz_g, in cm. Vertical scales differ between panels.

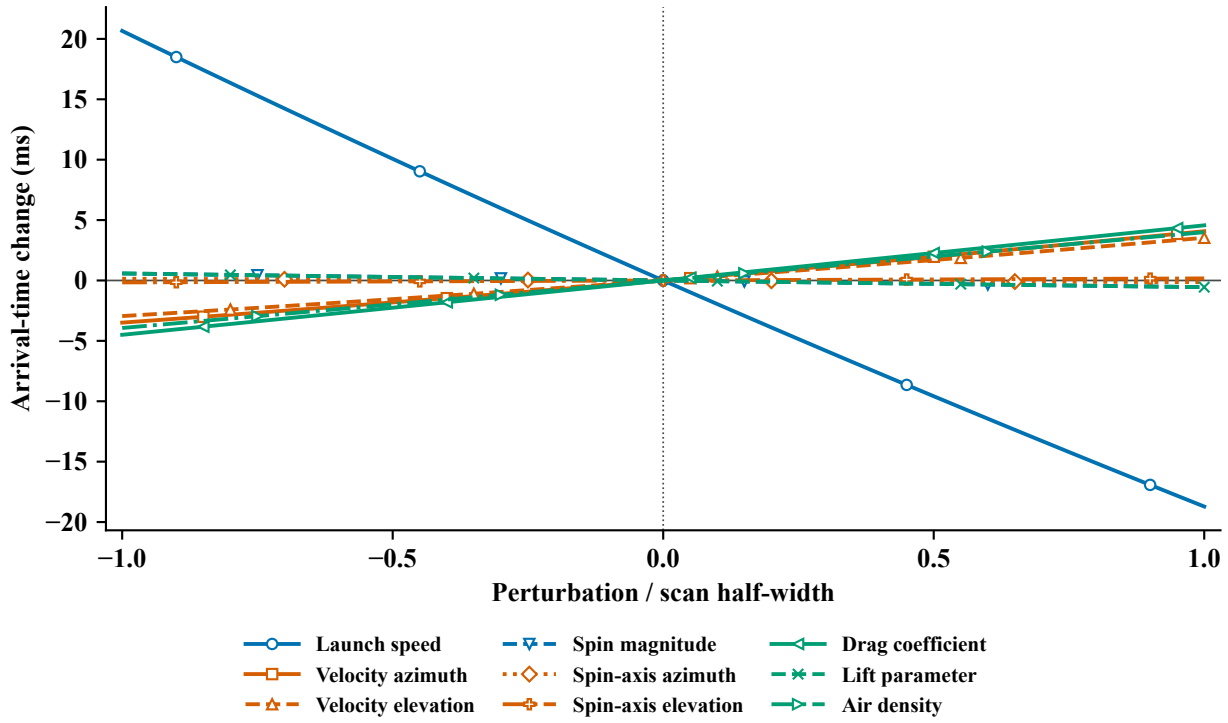


Figure 12: Arrival-time responses. Perturbation divided by scan half-width is plotted horizontally, so ± 1 denotes each scan endpoint. The vertical axis is Δt_g (ms); positive values mean later arrival.

5.2 Prediction Differences under Early-Position Constraints

The preceding analysis varies one parameter while holding the others fixed. We now examine how much the goal-plane crossing can change when different parameter combinations reproduce the same early positions approximately.

Using the aerodynamic reference values C_D^* and α^* from Section 4, define

$$\lambda_D = C_D/C_D^*, \qquad \lambda_\alpha = \alpha/\alpha^*. \tag{14}$$

Each coefficient multiplier takes 17 equally spaced values in $[1.2, 1.8]$, forming 17×17 combinations. For each combination, the initial spin component ω_{x0} takes 33 equally spaced values in $[-60, 60]$ rad/s. The three initial velocity components and the other two spin components are then fitted to the early positions in Table 4. The constraints are a total speed not exceeding 35 m/s, an absolute value not exceeding 60 rad/s for each spin component, and a maximum early-coordinate residual not exceeding 2 mm.

The supplementary search yields 5049 compatible trajectories across 153 aerodynamic-parameter combinations. Restricting both multipliers to $[1.35, 1.65]$, corresponding to $\pm 10\%$ changes in each aerodynamic parameter about its reference-trajectory value, gives 2079 trajectories across 63 combinations. Table 7 lists their crossing and arrival-time ranges, and Figure 13 shows their crossing positions.

Table 7: Goal-plane crossing ranges under early-position constraints

Range of each multiplier	y_g (m)	z_g (m)	t_g (s)
$[1.35, 1.65]$	2.5538–2.6550	1.9061–2.0369	0.3972–0.4025
$[1.2, 1.8]$	2.5478–2.6601	1.8972–2.0430	0.3972–0.4043

Within the wider parameter range, the largest crossing displacement among the searched trajectories is 0.1074 m relative to the reference. The minimum-residual solutions for each aerodynamic-parameter combination lie at the spin-component bounds; plotting these solutions alone produces two clusters. Intermediate spin states satisfying the same 2 mm position constraint form the band joining the clusters in Figure 13. Thus, initial velocity and spin can compensate for aerodynamic changes over the early observation interval, while different force histories accumulate into distinct subsequent crossings.

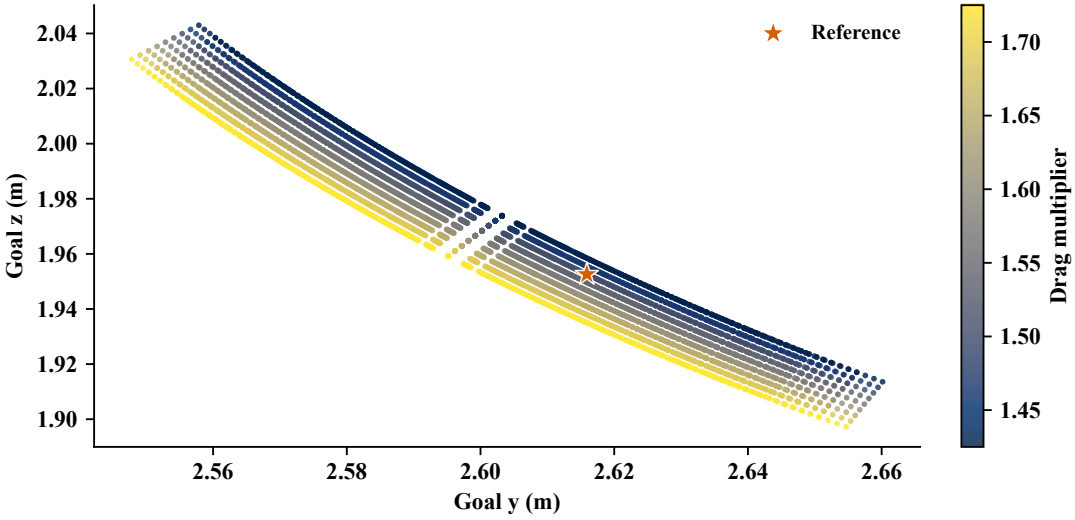


Figure 13: The 5049 crossings from the supplementary search under the 2 mm early-position constraint. Colour denotes λ_D; the orange star marks the reference trajectory.

5.3 Crossing Distribution under Joint Perturbations

To examine simultaneous changes in several conditions, the nine parameters are sampled independently and uniformly over the ranges in Section 5.1, producing 100000 inputs. Mass, radius and initial position remain fixed. Integrating the equations for each input gives the crossing distribution under jointly varying launch conditions and aerodynamic parameters.

Table 8: Crossings and arrival times for 100000 jointly perturbed trajectories

Output	Sample minimum–maximum	Central 95% interval, one coordinate
y_g (m)	1.9633–3.2143	2.1858–3.0505
z_g (m)	1.4381–2.4646	1.5534–2.3535
t_g (s)	0.3685–0.4346	0.3783–0.4218

The central 95% intervals in Table 8 use the 2.5th and 97.5th percentiles of each output. The mean crossing is $(2.6170, 1.9511)$ m, with lateral and vertical standard deviations of 0.2455 m and 0.2360 m. The largest displacement from the reference crossing is 0.801 m.

To describe lateral and vertical variation jointly, write $\boldsymbol{q}_i = (y_{g,i}, z_{g,i})^{\mathsf{T}}$ and use the sample mean $\bar{\boldsymbol{q}}$ and covariance matrix $\widehat{\Sigma}$ to define

$$d_i^2 = (\boldsymbol{q}_i - \bar{\boldsymbol{q}})^{\mathsf{T}} \widehat{\Sigma}^{-1} (\boldsymbol{q}_i - \bar{\boldsymbol{q}}). \tag{15}$$

Taking the empirical 95th percentile $c_{0.95} = 4.5916$ of d_i^2 gives the joint region

$$\mathcal{E}_{0.95} = \left\{ \boldsymbol{q} : (\boldsymbol{q} - \bar{\boldsymbol{q}})^{\mathsf{T}} \widehat{\Sigma}^{-1} (\boldsymbol{q} - \bar{\boldsymbol{q}}) \le c_{0.95} \right\}. \tag{16}$$

The set $\mathcal{E}_{0.95}$ is the ellipse boundary and its interior in Figure 14. The ellipse line marks the boundary, and the region contains 95% of the simulated crossings. The goal is 2.44 m high; accounting for the 0.11 m ball radius requires a centre height between 0.11 and 2.33 m and a lateral position satisfying $|y_g| \le 3.55$ m for the whole ball to pass through. By these criteria, 95772 trajectories clear the opening, or 95.772%; the other 4228 trajectories, or 4.228%, exceed the 2.33 m centre-height limit.

Figure 15 displays 400 randomly selected trajectories, linking the nine inputs to crossing coordinates and arrival time. Inputs are scaled to 0–1 using their perturbation bounds; outputs use the extrema of the full 100000-trajectory sample. The subset contains 385 blue lines for trajectories clearing the goal and 15 orange lines for trajectories crossing too high. Each line connects one set of launch conditions and aerodynamic parameters to its predicted outcome.

Figure 16 checks whether the crossing statistics stabilize as the number of simulations increases, informing the choice of 100000 samples for estimating the central 95% intervals. The blue sample extrema expand as additional extreme trajectories appear, while the orange percentile intervals gradually stabilize. Increasing the sample from 50000 to 100000 changes the lateral and vertical interval endpoints by at most 0.264 mm and 0.506 mm, and the arrival-time endpoints by at most 0.138 ms. Two additional runs of 10000 trajectories with independent random seeds differ from the main sample by at most 5.425 mm laterally, 4.320 mm vertically and 0.328 ms in time. These comparisons show that further sampling has a small effect on the central-interval estimates.

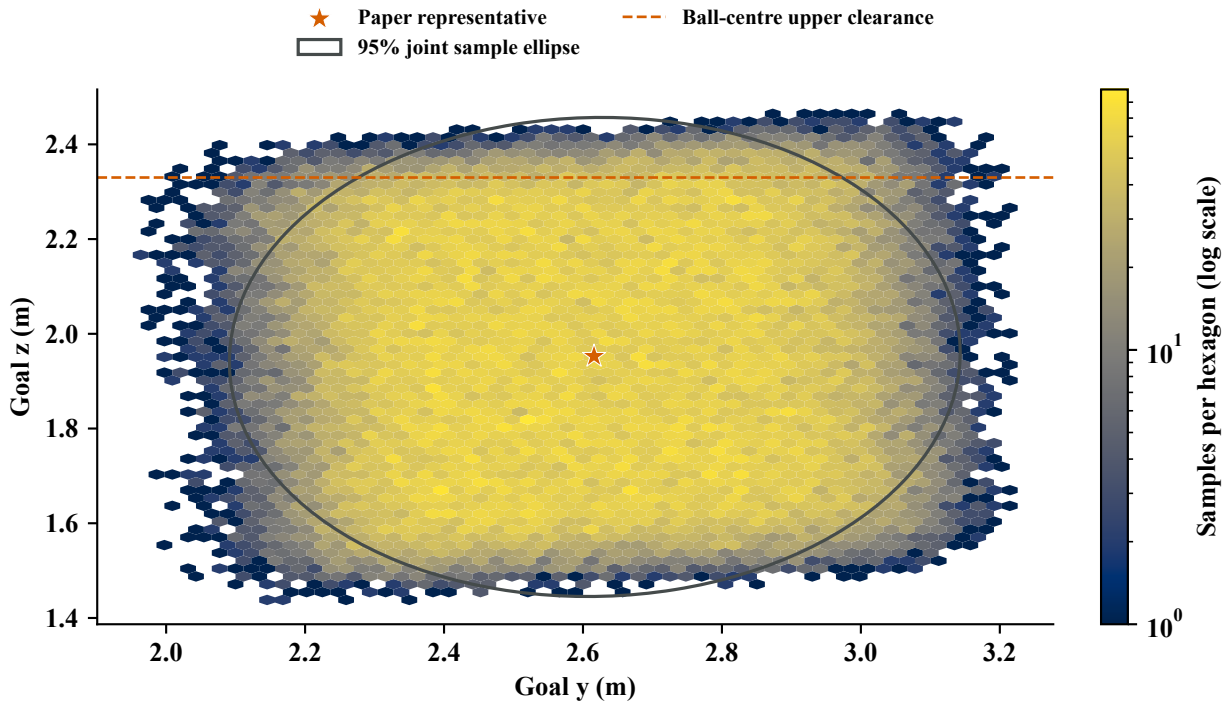


Figure 14: Goal-plane crossings under 100000 joint perturbations. Hexagon colour indicates sample count, the ellipse denotes the joint 95% region, and the dashed line marks the ball-centre height limit $z_g = 2.33$ m.

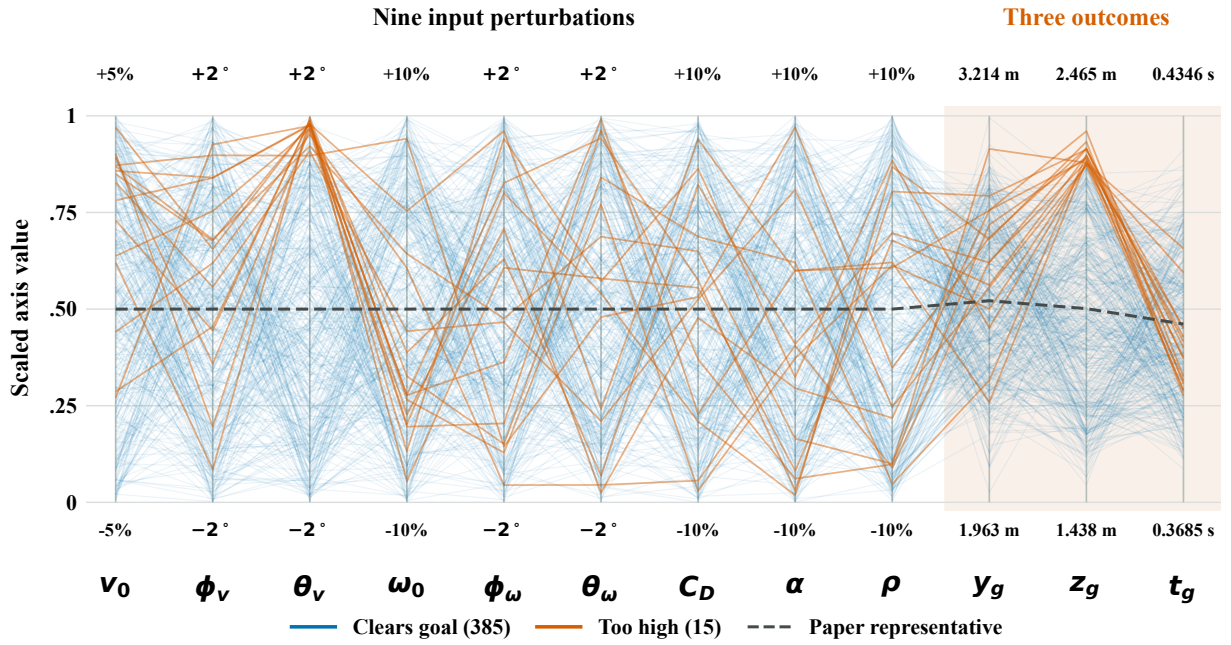


Figure 15: Parallel coordinates linking perturbed inputs, crossing coordinates and arrival time. The first nine axes are inputs and the last three are outputs. The displayed subset contains 385 trajectories clearing the opening and 15 crossing too high.

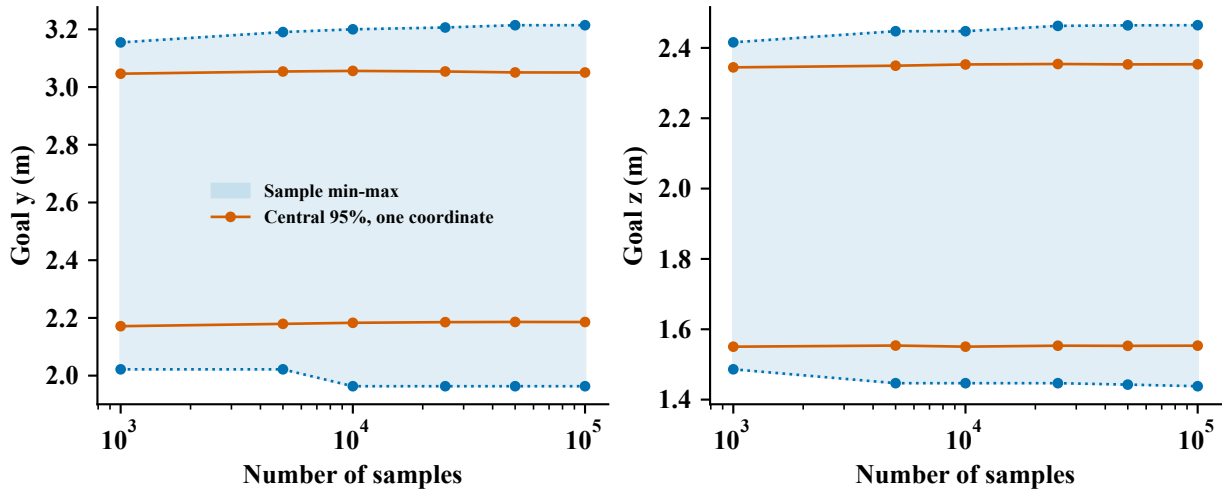


Figure 16: Convergence of crossing statistics with simulation sample size. Blue shading and dotted curves show sample extrema; orange curves show each coordinate's central 95% interval, illustrating the stability of interval estimates as sample size increases.

6 Discussion of Other Factors

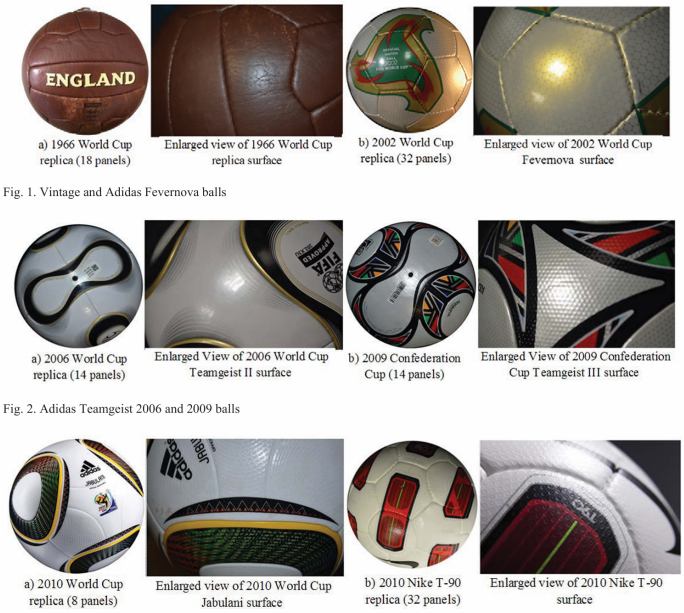
The main model isolates gravity, drag and Magnus lift. We now examine the physical effects omitted from its geometry, spin evolution and ambient airflow.

6.1 Asymmetric Surface Geometry

According to Paudel and Yan [4], full-flight simulations distinguish realistic football geometry from a smooth sphere. As shown in Figure 17, the comparison includes both idealized geometry and actual panel textures; as shown in Figure 17a, we distinguish the seamed and smooth geometries. For launch speeds of 10, 15 and 20 m/s and spin rates of 31.5, 63 and 94.5 rad/s, their results show that seams and surface structure alter separation, wake dynamics and Magnus force. The smooth sphere produces greater lateral bending in their nine comparisons, and vertical flight-time differences become more visible at higher speed and rotation.



(a) Illustrative seamed-football and smooth-sphere comparison, following Paudel and Yan [4].



(b) Six footballs and surface details, reproduced from Figures 1–3 of Alam et al. [10].

Figure 17: Football geometry, panels and surface textures.

According to Alam et al. [10], wind-tunnel measurements reveal different drag coefficients for six footballs and a smooth sphere over a range of Reynolds numbers. As shown in Figure 17b, these footballs have different panels and textures; as shown in Figure 18, their drag coefficients also differ. The differences illustrate the influence of ball surface structure on aerodynamic drag.

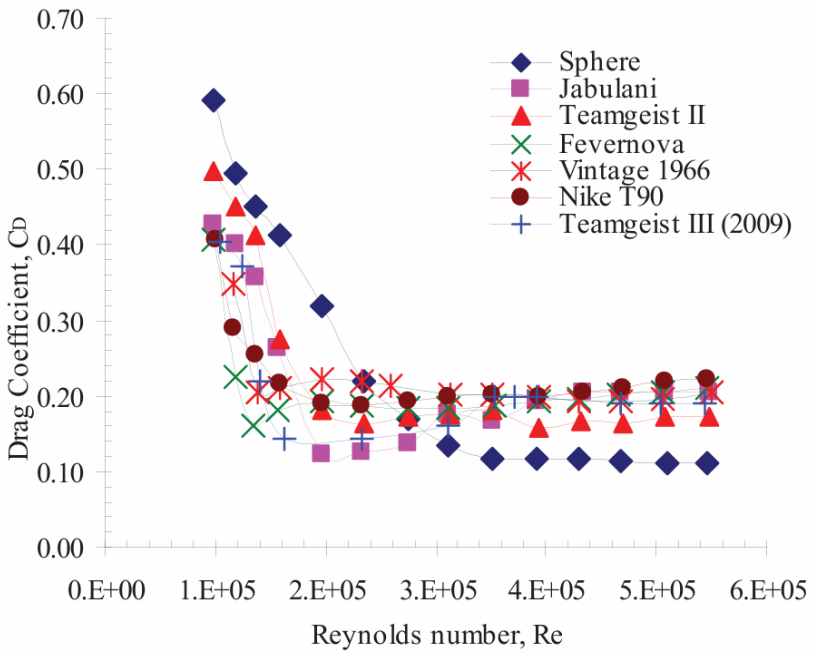


Figure 18: Drag coefficient against Reynolds number for different footballs and a smooth sphere, reproduced from Figure 5 of Alam et al. [10].

The spherical geometry in the present report describes the centre-of-mass motion and projected area. Its force coefficients come from football measurements, so the average surface influence enters through C_D and C_L. Explicit panel orientation and asymmetric wake fluctuations provide a further level of description when lateral-force details are important. These published results support retaining football-specific coefficients in a spherical trajectory model.

6.2 Rotation and Aerodynamic Damping

Aerodynamic torque opposes spin, gradually weakening rotation and changing the Magnus force. Following Grad et al. and the MuJoCo model [7, 8], we use high-Reynolds-number rotational damping: stronger spin produces stronger resistance. The dimensionless angular-drag parameter $C_{\text{angular}} = 1.5$ is the software reference value. Following Maponi and Piombin [3], the inertia is approximated by that of a thin spherical shell. Viscous damping is much weaker than the inertial contribution in this example, so the comparison focuses on the latter.

The reference curved shot has initial velocity $(23.6354, 0, 4.1676)$ m/s, spin $(0, 0, 35)$ rad/s and unit force corrections. Holding this state fixed, we compare constant spin with rotational damping. As shown in Figure 19, decreasing spin slightly changes the flight curvature and goal-plane crossing.

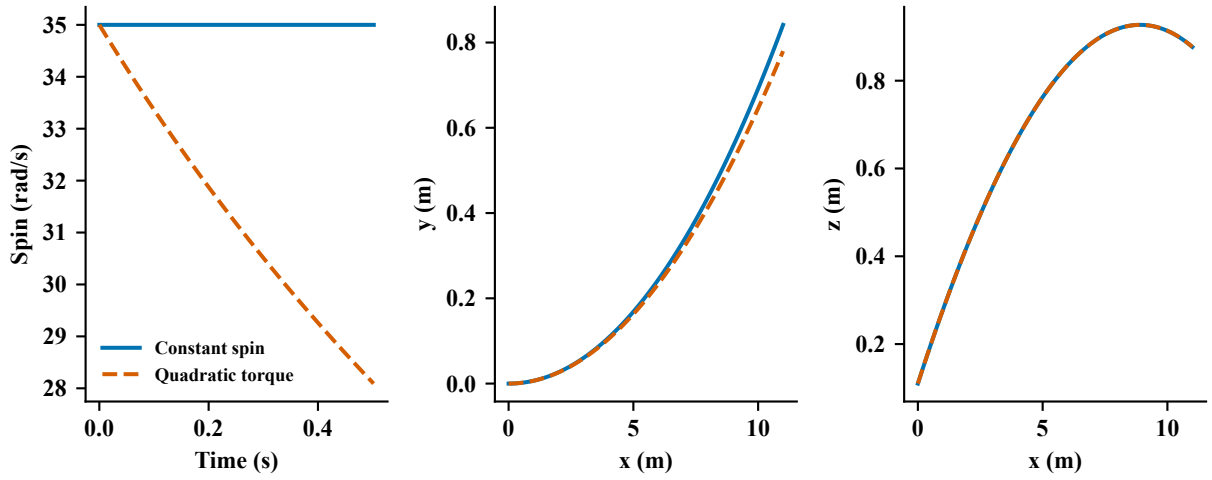


Figure 19: Constant spin and rotational damping under the same initial curved-shot conditions.

Spin falls by 19.740%, while the crossing moves 0.0616 m (0.560% of the 11 m distance) and arrival time changes by -0.066%. Rotational damping therefore has a small endpoint effect in this short reference flight.

6.3 Wind Speed and Force Estimates

Following the relative-airflow treatment of Maponi and Piombin [3], we introduce air-relative velocity $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{v}_{\text{wind}}$ in this subsection, where $\boldsymbol{v}_{\text{wind}}$ is air velocity and $u = \|\boldsymbol{u}\|$ is relative speed. For wind, $\boldsymbol{u}$ and u replace $\boldsymbol{v}$ and v in the aerodynamic relations of Section 3. Let w be wind speed in m/s and v the ball speed. At fixed coefficients, denote the still-air dynamic pressure by q_0 and the headwind, tailwind and crosswind values by q_{head}, q_{tail} and q_{cross}. From the squared relative-speed dependence,

$$\frac{q_{\text{head}}}{q_0} = (1 + \frac{w}{v})^2, \qquad \frac{q_{\text{tail}}}{q_0} = (1 - \frac{w}{v})^2, \qquad \frac{q_{\text{cross}}}{q_0} = 1 + (\frac{w}{v})^2. \tag{17}$$

Crosswind rotates the airflow by $\arctan(\frac{w}{v})$ and changes the force direction. As shown in Table 9, illustrative winds of 1, 3 and 5 m/s alter dynamic pressure at the 24 m/s curved-shot speed. Let $F_{L,0}$, $F_{L,\text{head}}$ and $F_{L,\text{tail}}$ denote lift magnitudes in still air, headwind and tailwind. From Eq. (4), the M4 linear lift law depends on the first power of relative speed. With fixed spin and $0 \leq w < v$,

$$\frac{F_{L,\text{head}}}{F_{L,0}} = 1 + \frac{w}{v}, \qquad \frac{F_{L,\text{tail}}}{F_{L,0}} = 1 - \frac{w}{v}. \tag{18}$$

Table 9: Illustrative wind at the 24 m/s curved-shot launch: dynamic-pressure changes at fixed coefficients.

w (m/s)	Headwind (%)	Tailwind (%)	Crosswind (%)	Angle (°)
1	+8.51	-8.16	+0.17	2.39
3	+26.56	-23.44	+1.56	7.13
5	+46.01	-37.33	+4.34	11.77

A 5% limit on the headwind dynamic-pressure change corresponds to $\frac{w}{v} \leq \sqrt{1.05} - 1$, or $w \leq 0.59$ m/s at this launch. The still-air approximation is most suitable for indoor or weak-wind flight; stronger wind should enter through the relative-velocity calculation.

7 Conclusions

The four models describe gravitational fall, deceleration, a constant resultant force and rotational deflection. M3 obtains constant acceleration from three positions. M4 inserts constant C_D and $C_L = \alpha S p$ into the motion equations, updating drag and Magnus force as speed changes.

For M4, the crossing errors are 0.360, 0.146 and 0.113 m for the three penalties. Its power-shot crossing error and subsequent path RMSE are the smallest among the four models. For placement 1, M3 passes through the early positions but reaches a maximum $x = 10.973$ m, short of the goal; M4 supplies a goal-plane prediction for this low flight.

In the competition-position example, a 1° launch-direction change moves the crossing by about 0.20 m, while speed strongly affects arrival time. Spin magnitude and α compensate through the linear lift law. The largest grid-search shift under the 2 mm position constraint is 0.1074 m; 95.772% of 100,000 jointly perturbed flights clear the opening.

Rotational damping shifts the reference curved-shot crossing by 0.0616 m. Wind enters through relative velocity: along the flight direction, drag changes with the squared speed ratio and linear Magnus force with the speed ratio.

8 Model Evaluation

8.1 Model Strengths

Connecting forces and trajectory changes. The model incorporates gravitational fall, drag-induced deceleration and rotational deflection in a common equation of motion, relating trajectory curvature to changes in force magnitude and direction. The four-model comparison shows how different force descriptions and the constant-acceleration approximation affect predictions, providing a physical interpretation of the differences between low placement shots and higher power shots.

Connecting early inference and later validation. Three early positions are used to estimate initial motion and relevant parameters, after which the corresponding equations predict the subsequent trajectory. Later measured positions provide comparisons of the path and goal-plane crossing, connecting early fitting with subsequent predictive validation. Shared coordinates and definitions of ground contact and goal-plane crossing also make results comparable across models and shot cases.

Reference predictions and crossing ranges. Individual-parameter scans distinguish the effects on crossing position and arrival time. The position-constrained search reveals differences in predictions from parameter combinations matching the early observations, while joint perturbations give crossing distributions over specified ranges. Together, these analyses connect parameter changes to the final prediction, describing a reference crossing, response trends and distribution ranges.

8.2 Model Weaknesses

Coupling of initial motion and aerodynamic parameters. Over a short observation interval, initial velocity, spin and aerodynamic parameters can compensate for one another, allowing initially similar trajectories to separate later in flight. In the linear lift relation, spin magnitude and the lift parameter enter the force through their product, so different parameter values can also generate the same trajectory. Additional early tracking positions and direct spin measurements would further constrain initial motion and aerodynamic parameters.

Simplified aerodynamics and environmental conditions. The main model treats the drag coefficient and linear lift parameter as constants during each flight and assumes still air and constant spin. Flow transitions, ball orientation and wake fluctuations alter aerodynamic forces, while wind and rotational damping affect relative airflow and spin evolution. More detailed descriptions require aerodynamic data for each ball type, together with wind and spin measurements.

Limited coverage of measured conditions. The measured comparison covers two placement shots and one power shot. Tracking records spanning additional ball types, shot techniques and wind conditions would allow assessment across more operating conditions. Joint perturbations use independent uniform sampling of nine parameters; estimating their distributions and correlations from measurements would bring the simulated crossing distribution closer to actual shooting conditions.

References

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[8] MuJoCo developers, MuJoCo 3.8.1, ellipsoid fluid model, `mj_viscousForces` in `src/engine/engine_passive.c`. Version 3.8.1 source; default angular drag: XML reference, geom/fluidcoef. Accessed 8 November 2026.

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[10] F. Alam, H. Chowdhury, M. Stemmer, Z. Wang and J. Yang, “Effects of surface structure on soccer ball aerodynamics,” *Procedia Engineering*, 34, 146–151, 2012. https://doi.org/10.1016/j.proeng.2012.04.026.

Appendix A: Data Sources, Frame Selection and Parameter Bounds

Following the dataset description by Bassek et al. [1] and the DFL release [2], we retain the source event and frame identifiers. The three selected penalties contain 39 records, with 15, 12 and 12 for the respective events. As shown in Table 10, each case is associated with its player, event identifier and selected samples.

Table 10: Correspondence between case labels and source records

Case	Event ID	Player	Selected samples
Placement 1	J03WMX	Dejan Ljubicic	3, 4, 5
Placement 2	J03WR9	Philipp Klement	3, 4, 5
Power shot	J03WOH	Dawid Igor Kownacki	2, 3, 4

The first selected position is the time origin and the fitting times are 0, 0.04 and 0.08 s. Their source-time origins are 0.08, 0.08 and 0.04 s in the table's order. Repeated and post-goal records are retained; later positions are used only for validation. The observed goal crossing uses a source frame or interpolation. Let $(X, Y, Z_{\text{original}})$ denote the published pitch-centred coordinates. The height is interpreted in metres as ball-centre height, and the local coordinates follow from $x = X - 41.5$, $y = Y$ and $z = Z_{\text{original}}$.

As shown in Table 11, the retained inputs and observed crossings share one local coordinate system. As shown in Table 12, the inverse variables have specified search ranges.

Table 11: Selected early positions and observed goal crossings in local coordinates.

Event	Source frame	t (s)	x (m)	y (m)	z (m)
Placement 1	152602	0.00	3.040	0.550	0.130
Placement 1	152603	0.04	4.190	0.820	0.150
Placement 1	152604	0.08	5.250	1.070	0.180
Placement 1 goal	152610	0.3200	11.0000	2.7500	0.1600
Placement 2	176245	0.00	2.570	0.850	0.340
Placement 2	176246	0.04	3.640	1.200	0.410
Placement 2	176247	0.08	4.710	1.530	0.470
Placement 2 goal	176253–176254	0.3295	11.0000	3.0697	0.4657
Power shot	128955	0.00	0.770	0.080	0.240
Power shot	128956	0.04	1.990	0.340	0.460
Power shot	128957	0.08	3.180	0.620	0.650
Power shot goal	128964–128965	0.3643	11.0000	2.9188	1.1789

Table 12: Component bounds and force-correction settings for inversion.

Quantity	Search interval
v_{x0} (M1, M2, M4)	$[0, 35]$ m/s
v_{y0}, v_{z0} (M1, M2, M4)	$[-35, 35]$ m/s
$\|\boldsymbol{\omega}_0\|$ (M4)	30 rad/s
Azimuth, elevation (M4)	$[-\pi, \pi], [-\frac{\pi}{2}, \frac{\pi}{2}]$ rad
$\frac{C_D}{C_D^*}, \frac{\alpha}{\alpha^*}$	$[0.5, 1.5]$

As shown in Table 13, the M4 states differ in speed, coefficients and unit spin axis $\boldsymbol{n} = \dfrac{\boldsymbol{\omega}}{\|\boldsymbol{\omega}\|}$.

Table 13: Fitted M4 states at the selected starting frames.

Event	v_0 (m/s)	C_D	α	n_x	n_y	n_z
J03WMX	29.20	0.3623	2.2877	0.2280	-0.9728	0.0413
J03WR9	28.34	0.1208	2.2877	0.1439	-0.2711	-0.9517
J03WOH	32.05	0.3238	2.2877	-0.2276	0.3170	0.9207

Reproduction Code

The following excerpts evaluate the force components and state derivatives used in Eq. (7). Initial-state inference and subsequent trajectory prediction use these same physical relations and coefficient settings.

```
def state_derivative(t, state, config):
    """The single nine-state RHS used for prediction and inverse fitting."""
    f = forces(state[3:6], state[6:9], config)
    return np.r_[state[3:6], sum(f.values())/config.mass,
                 rotational_torque(state[6:9], config)/config.moment_of_inertia]
```

```
def forces(velocity, spin, config):
    """Return force components on ball, in newtons."""
    v = np.asarray(velocity, dtype=float) - np.asarray(config.wind, dtype=float)
    omega = np.asarray(spin, dtype=float)
    speed = np.linalg.norm(v)
    weight = (config.mass * np.asarray(config.constant_acceleration, dtype=float)
              if config.constant_acceleration is not None else
              np.array([0., 0., -config.mass * config.gravity if config.use_gravity else
                  0.]))
    drag = np.zeros(3)
    magnus = np.zeros(3)
    if speed > 1e-12:
        cd, cl, _, _, _ = coefficients(velocity, spin, config)
        q_area = 0.5 * config.air_density * np.pi * config.radius**2 * speed**2
        if config.use_drag:
            drag = -q_area * config.drag_scale * cd * v / speed
        cross = np.cross(omega, v)
        norm = np.linalg.norm(cross)
        if config.use_magnus and norm > 1e-12:
            # The literature fit is for transverse spin. Project its effect
            # onto the plane perpendicular to airflow for an oblique spin axis.
            sin_angle = norm / (np.linalg.norm(omega) * speed)
            magnus = q_area * config.lift_scale * cl * sin_angle * cross / norm
    return {"gravity": weight, "drag": drag, "magnus": magnus}
```